

Lab 8 Worksheet — Wireless Standards, Channel Planning and Interference

Name: ____ Date: ____

Exam objective: Compare 802.11 standards and frequency bands, and design a channel plan that avoids co-channel and adjacent-channel interference (Core 1 objective 2.3).

Record as you go

Step	What you did	What you observed
1	Build the standards table with rows for 802.11a, b, g, n, ac and ax and columns for frequency band, maximum theoretical throughput, channel width and backward compatibility.	
2	Record that 2.4 GHz offers only three non-overlapping channels — 1, 6 and 11 — and explain from channel width why any other choice overlaps.	
3	Record that 5 GHz offers around 24 non-overlapping 20 MHz channels, and note the trade-off: more channels and more speed, but shorter range and poorer wall penetration.	
4	Open a Wi-Fi analyser on your device and record every visible SSID with its channel, band and signal strength in dBm.	
5	Identify from your scan which 2.4 GHz channels are most congested in your location and which of 1, 6 or 11 is least used.	
6	Interpret the signal strengths you recorded: better than -50 dBm is excellent, -60 is good, -70 is usable and worse than -80 is unreliable.	
7	Sketch a three-room office floor plan and place three access points to give overlapping coverage	

Step	What you did	What you observed
	with no dead zones.	
8	Assign 2.4 GHz channels 1, 6 and 11 to the three access points so that no two adjacent cells share a channel, and mark each on the sketch.	
9	Explain the difference between co-channel interference, where cells share a channel and must take turns, and adjacent-channel interference, where overlapping channels corrupt each other's transmissions.	
10	List five physical sources of 2.4 GHz interference in a typical office — microwave ovens, cordless phones, Bluetooth devices, fluorescent ballasts and thick or metal-reinforced walls.	
11	State the security configuration for all three access points: WPA3 where supported, WPA2 with AES-CCMP as the minimum, and never WEP or WPA with TKIP.	
12	Write the justification paragraph explaining why your channel assignment minimises interference, referencing the non-overlapping constraint explicitly.	

Verification

Success criterion: Your channel plan uses only channels 1, 6 and 11 in the 2.4 GHz band with no adjacent cells sharing a channel; the scan records real SSIDs with channel and dBm; and the justification correctly distinguishes co-channel from adjacent-channel interference.

- I completed every step in the lab.
- My result meets the success criterion above.
- I recorded my evidence (screenshots, output, completed tables).

Reflection

What surprised you in this lab?

Where would you apply this on the job?

What do you still need to revise before the exam?
